AI concepts for developers and technology professionals

1) Computer Vision

1. Computer vision is based on the manipulation and analysis of what kinds of values in an image?

Timestamps in photograph metadata

Pixels ✓

Image file names

2. What is the primary role of filters in a convolutional neural network (CNN) used for image classification?

To apply visual effects to enhance image appearance.

To extract numeric features from images for use in a neural network. ✓

To compress image size for faster processing.

3. Which description best matches a Vision Transformer (ViT?

A tool that applies filters to images, changing their visual appearance.

An AI agent that converts a large language model to a vision model.

A model that uses attention to process image patches and create contextual embeddings. ✓

2) Information extraction

1. Which of these statements accurately defines AI information extraction?

Using SQL code to query a database.

Copying files from cloud storage to local disks.

Analyzing unstructured content to identify and extract relevant fields and values. ✓

2. How is OCR used in the information extraction process?

Online Content Retrieval (OCR) is used to copy cloud data to a local device for processing.

Optical character recognition (OCR) is used to convert images of text into machine-readable text data. ✓

Open Conversion Routine (OCR) is a method for converting JSON data into images.

3. How does generative AI enhance the data extraction process?

By using semantic language models to accurately match extracted values to data fields. ✓

By manually coding extraction rules for each document type.

By generating new documents from scratch.

3) Generative AI & Agents

1. What is a large language model (LLM)?

A type of AI model designed to generate human-like text. ✓

A model that only processes images, not text.

A small, efficient model for mobile devices.

2. What is the purpose of tokenization?

To sort the words in a sentence alphabetically

To break down text into smaller units. ✓

To convert text into binary code for processing by computers.

3. What are embeddings?

Extra words added by a transformer model to enhance text generation.

Small language models used for specific tasks.

Vector-based representations of tokens that capture their semantic meaning. ✓

4. What does an attention layer do in a transformer model?

Removes irrelevant words from the input text.

Examines the relationships between each token and the tokens around it. ✓

Flags inappropriate content in the generated text.

5. What is the purpose of a system prompt?

To provide context and instructions to the AI model. ✓

To configure a generative AI model to run on a particular operating system.

To store user preferences for future interactions.

6. What is an agent in the context of AI?

An AI system that can perform tasks on behalf of a user. ✓

A generative AI model that operates in secret.

A human operator to whom generative AI models escalate requests they can't handle.

4) AI speech

1. What activity happens during the pre-processing stage of speech recognition?

The audio is converted to .wmv format.

Background noise is added to the audio signal.

Feature vectors are extracted from the audio waveform for modeling. ✓

2. What are phonemes?

Artifacts that are removed from the signal as part of the clean-up process.

The smallest unit of sound in speech. ✓

AI models that generate audio.

3. Why is it important to generate prosody in speech synthesis?

Prosody maximizes the volume of the audio output.

Prosody translates the speech to the language of the listener.

Prosody ensures natural pronunciation and speech cadence. ✓

5) NLP

1. What is the purpose of tokenization?

To translate text into another language.

To summarize large documents.

To break down text into smaller units for analysis. ✓

2. Which of the following techniques is used to determine the importance of words in a specific document within the context of a larger collection of documents?

Naïve Bayes

TF-IDF (Term Frequency-Inverse Document Frequency) ✓

Word2Vec

3. Which of the following best describes the role of embedding vectors in natural language processing (NLP)?

They duplicate tokens in multiple languages.

They define stopwords that should be ignored.

They capture semantic token relationships in multiple dimensions. ✓

6) Retrieval-augmented generation (RAG)

1. What is the main purpose of retrieval-augmented generation (RAG)?

To retrain a language model whenever source information changes.

To ground generated responses in relevant information retrieved from a data source. ✓

To replace a language model with a search engine.

2. Why are large documents commonly divided into chunks before they're indexed?

So the solution can retrieve focused passages that fit efficiently into a prompt. ✓

So every chunk can be used to train a separate language model.

So the original documents no longer need to be stored.

3. What does an embedding represent in a RAG solution?

A numerical vector that captures semantic characteristics of content. ✓

A set of instructions that defines the model's behavior.

A final answer generated for a user.

4. Which search approach can find a passage about 'company travel costs' when the query asks about 'business trip expenses,' even if the words differ?

Semantic vector search. ✓

An exact filename match.

Random document selection.

5. A RAG response is fluent but contains an incorrect policy limit. What should be examined first?

Whether the correct and current policy passage was retrieved and supplied as context. ✓

Whether the language model can produce a longer response.

Whether more unrelated documents can be added to the prompt.